Paper 003 · Reference Architecture for an AI Society

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Separate policy decisions from execution

An agent proposes an action. A policy decision point (PDP) evaluates identity, delegation, purpose, target, risk, configuration conformity, and revocation. A policy enforcement point (PEP) enforces the decision immediately before the operation. High-impact actions go through an execution adapter. Verification should cover potential paths that bypass the PEP.

Evidence connects intent, delegation and authority, the decision and policy version, the target and observed outcome, human intervention, incident handling, and recovery. Reconcile the authorization record with observations after execution, preserving gaps and discrepancies. Identity, execution, evidence, and oversight have separate logical responsibilities. Their deployment and independence are assessed for each implementation.

Traceability from requirements to verification

Paper 002 domain	Main control point	Verification question
ID	Identity and status resolution	Try a high-impact action with invalid identity and check denial
AUTH	Authority and policy decision/enforcement	Block an external action without authority (V-01)
DEL	Delegation validation	Test whether a child can exceed its parent's authority (V-02)
PUR	Purpose and target policy	Deny or escalate an out-of-purpose action
CON	Configuration and continuous conformity	Check that material change triggers reassessment (V-05)
EVD	Evidence	Reconstruct delegation, authority, decision, and outcome (V-04)
HUM	Human oversight	Test whether oversight can stop execution (V-06)
INC	Incident response	Check transition from detection to containment
REV	Revocation and pre-execution checks	Test whether parent revocation propagates and stops new operations (V-03)
REC	Recovery and reauthorization	Test whether old authority can return before remediation and review (V-07)
ACC	Role mapping and evidence	Distinguish who delegated, executed, and oversaw the operation

Paper 002 domain	Main control point	Verification question
INT	Interoperability gateway	Test whether unknown external credentials can become authority without validation (V-08)

V-01 through V-08 in this table are verification proposals in this paper. The technical proposal uses test identifiers within its own document.

Trust boundaries and a minimal proof of concept

Define what to recheck at delegation from a person or organization to an agent, redelegation, transfer of permission to the execution point, action on an external target, exchange of evidence, and acceptance of credentials across organizations. A minimal proof of concept can cover seven logical functions: identity, authority, decision and enforcement, tool access, evidence, oversight, and revocation and recovery.

Check scope, configuration changes, and revocation at runtime. Restrict or suspend operations when needed. After remediation, reassessment, and review, reissue only the authority required for recovery.

Scope of verification

This is a reference design. A first evaluation would cover allowed and denied actions, missing evidence, material change, revocation, suspension, and recovery in a simulated environment, and report what could be observed. These are proposed tests. This paper presents no measured results for an integrated implementation.

Related research turns enforcement and independent observation into a concrete file-update test plan. Verifying AI Agent Authorization and Execution Evidence

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